

The population generative model

1 What the model does

The model gives a distribution of parameters for a population of PDAC patients. One draw from this distribution is one patient. The model then predicts what a published study of these patients shows. The fit compares this prediction with the numbers in the papers.

We never observe one patient. We observe a median and an interquartile range from a cohort. The model must therefore run all the way from a population of parameters to a cohort summary before it can compare anything.

2 The terms

Throughout, $\odot$ is an elementwise product and $\text{diag}(v)$ is the diagonal matrix with v on its diagonal. A logarithm of a vector acts elementwise.

Each table gives the status of the term. **inferred** means that the fit estimates it. **stage 1** means that we compute it from the stage-1 output before the fit. **input** means that we set or build it before the fit by some other route. **derived** means that it is a function of the other terms. **data** means that we read it from a paper. Six quantities are **inferred**; everything else is not.

Sizes

All sizes are constants.

Term	Value	Meaning
P	271	The number of model parameters.
M	65	The number of observables, over all scenarios.
N	2000	The number of patients that we simulate.
C	27	The number of study cohorts.
R	163	The number of anchor rows (one quantile of one observable).
n_c	6–702	The number of real patients in cohort c .

Quantities for one patient

Term	Size	Status	Meaning
θ_i	P	derived	The parameter vector of patient i . All values are positive.
$\vartheta_i = \log \theta_i$	P	derived	The same vector on the log scale.
z_i	P	input	The latent coordinates of patient i .
x_i	M	derived	The observables of patient i , on the log scale.

Quantities for the population

Term	Size	Status	Meaning
μ	P	inferred	The centre of the population, on the log scale.
ω	P	inferred	The spread of each parameter between patients, on the log scale.
ω_0	P	input	The prior centre for ω . Section 7 builds it in four layers. Layer 4 uses stage-1 output for 9 parameters.
Ω	$P \times P$	derived	The covariance of the population, on the log scale.
R	$P \times P$	stage 1	The correlation matrix of the population. Equation (18) computes it from the stage-1 posterior copula. That copula covers 65 of the 271 parameters; R is the identity elsewhere.
L_R	$P \times P$	input	The Cholesky factor of R , so that $R = L_R L_R^\top$.
D_ω	$P \times P$	derived	$\text{diag}(\omega)$, the matrix of marginal widths.

Quantities for the data

Term	Size	Status	Meaning
g	map	input	The model. It changes log parameters into log observables.
$Q(\phi)$	R	derived	The quantiles that the model predicts.
y	R	data	The quantiles that the papers report.
η	C	inferred	An offset for each study.
δ	M	inferred	A structural error in the centre, for each observable.
κ	M	inferred	A structural error in the width, for each observable.
κ_0	1	inferred	The width error shared by every observable.
ε	M	input	The held-out residual of the emulator, for each observable.
Σ	$R \times R$	input	The covariance of the observed quantiles. The fit does not change it.

Quantities that control the priors

Term	Value	Status	Meaning
μ_0	P	stage 1	The centre of the prior on μ . The mean of the stage-1 prior, on the log scale.
τ_μ	0.5	input	The width of the prior on μ .
τ_ω	0.5	input	The width of the prior on ω , on the log scale.
τ_η	—	inferred	The scale of the study offsets.
τ_δ	—	inferred	The scale of the centre errors.
τ_κ	—	inferred	The scale of the width errors.
ϕ	—	inferred	All the quantities that the fit estimates.

The full parameter vector is $\phi = (\mu, \log \omega, \eta_{\text{raw}}, \delta_{\text{raw}}, \kappa_{\text{raw}}, \log \kappa_0, \log \tau_\eta, \log \tau_\delta, \log \tau_\kappa)$.

3 How to make one patient

1. Draw z_i from a standard normal distribution. Do this one time only, before the fit.
2. Multiply by L_R . This step installs the correlations.
3. Multiply element j of the result by ω_j . This step sets the width of each parameter.
4. Add μ .
5. The result is ϑ_i .
6. Take the exponential of each element. The result is θ_i .

In one equation:

$$\vartheta_i = \mu + D_\omega L_R z_i, \quad z_i \sim \mathcal{N}(0, I_P), \quad \theta_i = \exp(\vartheta_i). \quad (1)$$

The covariance of the population follows directly:

$$\vartheta_i \sim \mathcal{N}(\mu, \Omega), \quad \Omega = D_\omega L_R L_R^\top D_\omega = D_\omega R D_\omega. \quad (2)$$

This form separates the two halves of a covariance matrix. R holds the correlations and the fit does not change it. ω holds the marginal widths and the fit estimates all P of them. The construction is known as the separation strategy.

Two properties follow, and both matter for interpretation.

$$\sqrt{\Omega_{jj}} = \omega_j \quad \omega_j \text{ is exactly the width of parameter } j. \quad (3)$$

$$\text{Corr}(\Omega) = R \quad \text{The correlations do not move when the fit moves } \omega. \quad (4)$$

A diagonal rescaling cannot change a correlation matrix, so the second property holds at every value of ω , not only at one point.

Why step 1 happens only one time. We draw z once and we keep it. The map from ϕ to $Q(\phi)$ is then smooth and has no random noise. The sampler needs this property to calculate a gradient. The same N patients move continuously as ϕ moves. The cost is a fixed approximation error of order $N^{-1/2}$ in Q , instead of an error that averages away. Section 5 adds this error to Σ .

4 How to predict the data

1. Make N patients with the steps in the previous section.
2. Put each patient through the model g . This gives N observable vectors $x_i = g(\vartheta_i)$.
3. For each observable, sort the N values.
4. Read the values at the probability levels 0.25, 0.50 and 0.75.
5. Collect these values into $Q(\phi)$.

Written as an equation, the prediction for observable m at level p is

$$Q_m(p; \phi) = \hat{F}_{m,\phi}^{-1}(p), \quad \hat{F}_{m,\phi}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{x_{im} \leq t\}. \quad (5)$$

There is no closed form for $F_{m,\phi}$. The model g is a set of differential equations and it is not linear, so the pushforward of a log-normal through it belongs to no standard family. Step 3 is therefore a numerical calculation, not a shortcut. In practice g is a neural-network surrogate for the simulator, which is what makes $N = 2000$ evaluations per gradient step affordable.

The levels 0.25, 0.50 and 0.75 are what the sources report. All 65 targets use this grid.

5 The observation equation

The model corrects its prediction in two ways before the comparison. One correction moves the centre. The other changes the width.

$$\tilde{Q}_m(p) = \underbrace{Q_m(0.5) + \kappa_m(Q_m(p) - Q_m(0.5))}_{\text{width correction}} + \underbrace{\eta_{c(m)} + \delta_m}_{\text{centre correction}}. \quad (6)$$

Then

$$y = \tilde{Q}(\phi) + e, \quad e \sim \mathcal{N}(0, \Sigma). \quad (7)$$

Each term answers one question about how the model can be wrong.

η_c absorbs a measurement offset that applies to a whole study. δ_m absorbs a systematic error of the model in the centre of observable m . κ_m absorbs a systematic error of the model in the *width* of observable m : it stretches the predicted quantiles about their own median, so $\kappa_m = 1$ means no error and $\kappa_m = 2$ means the model produces half the spread that the study reports.

Without κ the model has no way to be wrong about a width. Only ω could then close a gap in spread, and the fit would report a fault of the simulator as diversity between patients. Section 6.4 explains what separates the two.

Σ is the covariance of a quantile that comes from n_c patients. For two rows of the same cohort, at observables j, l and levels p_a, p_b :

$$\Sigma_{(j,p_a),(l,p_b)} = \frac{C_{jl}(p_a, p_b) - p_a p_b}{n_c f_j(Q_j(p_a)) f_l(Q_l(p_b))}. \quad (8)$$

In this equation:

- p_a and p_b are the two probability levels.
- n_c is the number of patients in the cohort.
- f_j and f_l are the densities of the two observables at those quantiles. We estimate them by quantile spacing, $f_m(Q_m(p)) \approx 2h / \{Q_m(p+h) - Q_m(p-h)\}$ with $h = 0.035$.
- C_{jl} is a joint probability, given below.

If the two rows are for the same observable, then

$$C_{jj}(p_a, p_b) = \min(p_a, p_b). \quad (9)$$

This value is exact.

If the two rows are for two different observables, then the model uses a Gaussian copula:

$$C_{jl}(p_a, p_b) = \Phi_2(z_{p_a}, z_{p_b}; \rho_{jl}), \quad z_p = \Phi^{-1}(p), \quad (10)$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate normal distribution function and ρ_{jl} is the normal-score correlation of the two observables across the cloud. Rows from different cohorts have a covariance of 0.

Two further errors go on the diagonal of (8). The first is the error of the emulator, ε_m , which is its held-out residual standard deviation for observable m . The second is the Monte Carlo error of Q , which comes from the finite cloud. We fold the second into the sample size, because a

quantile from n_c real patients compared against a quantile from N simulated patients behaves like one from n_{eff} patients:

$$\frac{1}{n_{\text{eff}}} = \frac{1}{n_c} + \frac{1}{N}, \quad \Sigma \leftarrow \Sigma|_{n_c \rightarrow n_{\text{eff}}} + \text{diag}(\epsilon^2). \quad (11)$$

Σ does not change during the fit. The terms f_m and ρ_{jl} depend on the population, so they depend on ϕ . We evaluate them once, at a plug-in value ϕ_0 , and then hold them fixed. The method is therefore an estimating equation, not a full likelihood.

This is deliberate. If Σ changes with ϕ , then Σ grows with σ_u^2 , and the term $-\frac{1}{2} \log \det \Sigma$ rewards a narrow population whatever the fit quality. A test showed the size of the error. With cohorts of 20, 40 and 120 patients, a live Σ recovered 0.878, 0.941 and 0.963 of the true spread. A fixed Σ recovered 1.002, 0.999 and 1.001.

The plug-in value is not critical. A change of ϕ_0 by a factor of two moved the estimate by 0.14 to 0.32 sampling standard deviations.

6 The priors

6.1 The centre

$$\mu_j \sim \mathcal{N}(\mu_{0j}, \tau_\mu^2), \quad j = 1, \dots, P, \quad \tau_\mu = 0.5. \quad (12)$$

The vector μ_0 is the mean of the stage-1 composite prior on the log scale. We estimate it from 2×10^4 draws.

The value τ_μ measures how well we know the centre of the population. It shrinks as data accumulate. It is a different quantity from ω , which is the variability between patients and does not shrink. To confuse the two is the standard-error-versus-standard-deviation error, one level up in the hierarchy.

Known limitation. The value τ_μ is one number for every parameter. A parameter that stage 1 determined to within 0.05 and a parameter constrained only to within 1.5 therefore get the same width. The stage-1 correlations do not enter the prior on μ at all. The per-parameter widths s_j are available and unused. The principled choice is to replace $\tau_\mu^2 I_P$ by the stage-1 covariance itself.

6.2 The spread

$$\log \omega_j \sim \mathcal{N}(\log \omega_{0j}, \tau_\omega^2), \quad j = 1, \dots, P, \quad \tau_\omega = 0.5. \quad (13)$$

The centre ω_0 comes from the four-layer construction in section 7.

The value τ_ω controls how far the data can move the spread away from its anchor. On the log scale, $\tau_\omega = 0.5$ gives a central 95% interval of about $[0.37, 2.72]$ for each parameter.

This choice is not harmless. The data constrain at most about 50 functions of Ω (section 9), against $P = 271$ free values of ω . Most parameters therefore return their prior. A large τ_ω reports a wide population that no study measured.

6.3 The corrections

$$\eta = \tau_\eta \eta_{\text{raw}}, \quad \eta_{\text{raw}} \sim \mathcal{N}(0, I_C), \quad \log \tau_\eta \sim \mathcal{N}(\log 0.15, 1), \quad (14)$$

$$\delta = \tau_\delta \delta_{\text{raw}}, \quad \delta_{\text{raw}} \sim \mathcal{N}(0, I_M), \quad \log \tau_\delta \sim \mathcal{N}(\log 0.15, 1), \quad (15)$$

$$\log \kappa = \log \kappa_0 + \tau_\kappa \kappa_{\text{raw}}, \quad \kappa_{\text{raw}} \sim \mathcal{N}(0, I_M), \quad \log \tau_\kappa \sim \mathcal{N}(\log 0.15, 1), \quad (16)$$

with $\log \kappa_0 \sim \mathcal{N}(0, 0.5^2)$. These are non-centred forms. The centre 0.15 matches the floor of the translation-relevance rubric used at stage 1.

The width correction splits into two parts on purpose. κ_0 is one number for every observable. κ_m then departs from it by τ_κ . Section 6.4 uses that split.

6.4 What separates a narrow model from a narrow population

Both κ and ω can widen a predicted interquartile range, so we must say what tells them apart.

κ_0 is one number that stretches every observable by the same factor. ω has P entries, and each one widens a different set of observables, through the sensitivities J_m . A uniform shortfall in width is therefore cheap under κ_0 and expensive under ω , because no single parameter feeds every observable. A shortfall with structure is the reverse.

The current model produces about 0.62 of the observed interquartile range at the median observable. Under this form, that uniform part goes to κ_0 , and ω moves only for the pattern that remains.

Per observable, the split is not identified. Each observable has three anchor rows. δ_m takes the centre and κ_m takes the width, which leaves one degree of freedom. So τ_δ and τ_κ carry the separation, exactly as in section 10.

7 How we build ω

The vector ω_0 is the prior centre for the marginal widths in (13). We build it in four layers. Each layer replaces the layer before it.

- L1. Default.** $\omega_j = 0.35$ for every parameter. This is a coefficient of variation of about 36%. A parameter gets this value when we know nothing more specific.
- L2. Class.** A geometric packing limit gets 0.15. A material constant gets 0.10. The spread that we observe for these classes is mostly assay noise. To call it ω would overstate the real diversity.
- L3. Explicit.** One value for one parameter, when we hold a specific belief about it.
- L4. Data.** A submodel target can report a spread across biological units. We then combine that spread with the value from L1 to L3. We do not replace it.

For L4, the sampling standard deviation of a log spread from n units is about $\{2(n-1)\}^{-1/2}$. The data therefore carry precision $2(n-1)$ against a class-prior precision τ^{-2} . This gives

$$\log \omega_j = \frac{2(n_j - 1) \log \hat{s}_j + \tau^{-2} \log \omega_j^{(0)}}{2(n_j - 1) + \tau^{-2}}, \quad \tau = 0.5. \quad (17)$$

In this equation, $\hat{s}_j$ is the spread that the source reports, n_j is the number of biological units behind it, and $\omega_j^{(0)}$ is the value from L1 to L3. The data become stronger than the prior at about $n = 3$.

A parameter with no value of n_j keeps $\omega_j^{(0)}$. A spread with no sample size carries no information.

What L4 admits. Equation (17) applies only when the source measures variation across biological units, such as patients, donors or animals. It does not apply to technical replicates, to translation between experimental contexts, or to an assumed value. Those are measurement or epistemic quantities and not variability between patients. This gate matters more than it appears. The submodel population block covers 178 parameters, but for 145 of them the reported spread is numerically equal to the epistemic posterior width. Without the gate, the model would take those 145 epistemic widths as between-patient spread. At present the gate admits 9 of 271 parameters. The other 262 come from L1 to L3.

8 Where R comes from

R is a separate input to the model. Nothing in the fit changes it, and you can replace it without touching anything else. At present it is the correlation matrix of the stage-1 composite prior,

$$R = \text{Corr}(\text{Cov}_\pi[\theta]). \quad (18)$$

The stage-1 prior joins submodel posteriors through a Gaussian copula, so (18) carries whatever structure that fit produced. The copula covers 65 of the 271 parameters. The remaining 206 parameters come from independent marginals, so R states independence for every pair that involves one of them.

What (18) does and does not give. Two measured facts qualify it. The matrix is close to the identity. Its copula covers 65 of 271 parameters, and only 26 off-diagonal entries exceed 0.5 in absolute value, out of 36,585. For most parameter pairs, (18) states independence. Where the matrix is not close to the identity, the entry is epistemic. The largest, at -0.994 , joins two rates that one target constrains only through their sum. That number describes what the stage-1 data could not separate. It does not describe how patients differ. A second group of strong entries comes from declared hierarchical parameter groups; those encode a real claim that the members are the same kind of quantity, but their magnitudes are still epistemic. R is therefore best read as a placeholder with the right interface, not as a measured quantity.

Better candidates for R exist and slot into the same place:

- a block matrix built from the declared parameter groups, with a within-group correlation elicited rather than read from stage 1;
- structural constraints, such as rates that partition a conserved total;
- the identity, if the conclusion is that nothing is known;
- a shrunk form $(1 - \lambda)I + \lambda \text{Corr}(\text{Cov}_\pi[\theta])$.

Because R enters only through (1), a comparison across these choices is a sensitivity analysis and not a redesign. Section 12 records that this comparison has not yet been run.

9 What the data constrain

The data constrain Ω only through the width of an observable distribution. To first order, the spread of observable m across patients is $J_m \Omega J_m^\top$, where $J_m = \partial g_m / \partial \vartheta$. Each observable that reports a scale therefore gives one linear function of Ω .

Published summaries are marginal, one observable at a time. The cross terms $J_j \Omega J_l^\top$ are never observed. Of the 65 targets, 49 report a usable scale, and every one uses the same three probability levels. At most about 50 functions of Ω are therefore identified.

Two consequences follow.

First, no part of R can be estimated from these data. Its entries are exactly the cross terms that a marginal summary never reports. R must come from outside the fit, which is why section 8 treats it as an input.

Second, most of the 271 widths are not informed either. Along those directions the posterior returns the prior (13). The reported population is then a statement of ω_0 , not a measurement.

Design note: there is no basis matrix. An earlier version replaced (1) by $\vartheta_i = \mu + W(\sigma_u \odot z_i)$. Here $W = L\tilde{V}$ came from an eigendecomposition of the observation information in the metric of the prior covariance, and the fit estimated $K = 15$ direction multipliers instead of 271 widths. The construction is known as the likelihood-informed subspace, or the active subspace.

The rotation and the truncation were removed. The correlations were kept, and they are now R .

Three reasons. First, σ_u was not interpretable: it scaled directions, so no single entry was the width of any parameter. Second, the correlations did not stay fixed. Because $\Omega = \Gamma + \sum_{k \leq K} (\sigma_{u,k}^2 - 1) w_k w_k^\top$, the fit moved $\text{Corr}(\Omega)$ by a rank- K update, and that update lay along the directions that the *measurements* resolve. A correlation that appears because a combination happened to be observable is not a biological statement. Third, the truncation to K directions limited how much model error the fit could report as patient diversity. That is protection by restriction and not by correctness; assumption (iv) names the correct fix.

The present form keeps what the rotation was supposed to deliver, namely the prior correlations, and states them as an explicit and replaceable object.

10 The corrections are aliased

Every observable belongs to exactly one cohort. We measured this: all 65 observables appear in one cohort each, and none appears twice. That fact has a consequence that the notation hides.

Take a cohort c that holds observables $m_1, \dots, m_k$. Equation (6) uses only the sum $\eta_c + \delta_{m_i}$. Add a constant to η_c , subtract the same constant from every δ_{m_i} in that cohort, and the likelihood does not change. η and δ are therefore aliased, exactly, and only their priors separate them.

What η is, and is not. In an ordinary meta-analysis, a study effect is identified because the same endpoint appears in several studies. No endpoint appears twice here. η_c is therefore not an identified study effect.

Read it instead as a correlation structure on the corrections: the model states that the

errors within one study share a common part. That claim has content, and it is a reasonable claim, but it is a prior structure and not a measurement.

The count is worth stating plainly. The model has 65 values of δ , 65 of κ and 27 of η , against $R = 163$ anchor rows. Sixteen observables have a single anchor row each, because their source reports a centre only; for those, δ_m absorbs the residual exactly.

Therefore μ is not identified by any single observable. It is identified by the *agreement* of residuals across observables, through the shrinkage of δ toward zero. The same holds for ω through κ . The scales τ_δ and τ_κ do the work, and they are estimated from how much the corrections disagree with each other.

This is a partial-pooling argument and it is sound. But the model does not have 163 independent measurements of the population, and no report should suggest that it does.

11 What the fit estimates

Quantity	Status	Source
μ	estimated, P	Prior centre from stage 1. Width $\tau_\mu = 0.5$.
ω	estimated, P	Prior centre ω_0 . Width $\tau_\omega = 0.5$.
η, δ	estimated	Centre corrections. Aliased; see section 10.
κ, κ_0	estimated	Width corrections. See section 6.4.
$\tau_\eta, \tau_\delta, \tau_\kappa$	estimated	These carry the separation of the corrections.
ε	fixed	Emulator residual, on the diagonal of Σ .
$R = \text{Corr}(\Omega)$	fixed	Section 8. An input, not a result.
ω_0	fixed	Section 7.
Σ	fixed	Equation (8), at a plug-in ϕ_0 .
$z_{1:N}$	fixed	Drawn one time.

The fit estimates $2P + C + 2M + 6$ quantities against $R = 163$ anchor rows. Only about 50 functions of Ω are identified by the data, so the prior carries most of ω , and the shrinkage of the corrections carries the rest.

12 Assumptions

- (i) **We assume R rather than estimating it.** The correlations of the population are an input (18), and no part of them is identified by these data. The current choice is the stage-1 correlation matrix, which is epistemic: it shows how our uncertainties move together, not how patients differ. It is also close to the identity. Section 8 lists better candidates. The sensitivity of the fit across those candidates has not been measured, and it should be, since R is assumed rather than known.
- (ii) **No data gives the correlations between patients.** Papers report each observable on its own, so the cross terms $J_j \Omega J_l^\top$ are never observed. To measure them, we need two or more observables in the same individuals. Digitised per-patient figures could supply this, but only if the extraction keeps the identity of each patient across panels.
- (iii) **The spread is mostly prior, not data.** About 50 functions of Ω are identified against 271 free widths. Also, 262 of the 271 values of ω_0 come from a class default rather than a measurement. The width of the reported population is therefore in large part a statement of assumption. This is why ω_0 must come from external evidence, and why τ_ω should not be widened for convenience.

- (iv) **The width correction and the population spread compete.** The term κ exists so that a narrow simulator does not have to be reported as diverse patients. Section 6.4 argues that κ_0 takes the uniform part and ω takes the structured part. That argument rests on no single parameter feeding every observable, which is true but is not a proof. Check it: fit synthetic data generated with a known ω and a known uniform narrowing, and confirm that the fit puts each in the right place.
- (v) **The corrections are aliased and the priors separate them.** Section 10 shows that η_c and δ_m are exactly aliased, because no observable appears in two cohorts. The count of anchor rows overstates the information by a wide margin. Any statement about μ or ω depends on τ_δ and τ_κ remaining small, which is an assumption about how consistent the sources are.
- (vi) **The emulator error term exists but is not populated.** The slot ε_m is in the model and enters Σ through (11). In the current pipeline nothing writes to it, so it is zero. The emulator's held-out residual is not negligible: relative to the within-cohort spread it lies between 0.095 and 0.254, and the range is wide because the pool used to measure the denominator does not vary most parameters. Until ε is wired, every target is trusted more than it should be, and the highest- n targets most of all.
- (vii) **The measured spreads are biased in both directions.** A reported across-patient spread also contains assay noise and biopsy-site variation, which makes it too large. A spread measured across animals of one genetic background is smaller than a spread across human patients, which makes it too small. We correct neither effect in (17).